



The illustration depicts a spacious, modern home goods store. The interior is filled with tall wooden shelves stocked with various products, primarily in white and blue packaging. Several customers are shown in different stages of shopping. In the foreground on the left, a young man in a blue jacket examines a product on a shelf, holding a blue tote bag that says "Good things inside." In the center, a woman in a tan coat walks away from the viewer. To the right, a man with a backpack and a woman with a large tote bag that says "Love Where You Live." are looking at products. A brown paper shopping bag with the text "Live Well. Shop Smart." is on the floor near them. A central display table holds various small containers and plants. A sign on the table reads "Live Well. Shop Smart." Above the store, a large sign says "Made for living." and various section signs like "Kitchen & Dining", "Bath & Body", and "Storage & Organize" are visible. A sign on the left wall says "Less clutter, more life." and another in the background says "For the way you live."

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Dataset Size

Rows: 3,900

Columns: 18

Missing Data: 37 values in Review Rating column

Key Features

- **Customer demographics** (Age, Gender, Location, Subscription Status)
- **Purchase details** (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- **Shopping behavior** (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender

Compared total revenue generated by male vs. female customers.

gener	revenue
Female	75191
Male	157890

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

custmer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88

Total rows: 839

Query complete 00:00:01

3. Top 5 Products by Rating

Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Data Analysis using SQL (Business Transactions)

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

shipping_type	round
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

6. Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

4. Data Analysis using SQL (Business Transactions)

7. Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number of Customers
Loyal	3116
New	83
Returning	701

8. Top 3 Products per Category

Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions

Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
No	2518
Yes	958

10. Revenue by Age Group

Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



Customer Behavior Dashboard

Filters

Subscription Status

No Yes

Gender

Female Male

Category

Accessories

Clothing

Footwear

Outerwear

Shipping Type

2-Day Shipping

Express

Free Shipping

Next Day Air

Standard

Store Pickup

Key Metrics

3.9K

Number of Customers

\$59.76

Average Purchase Amount

3.75

Average Review Rating

% of Customers by Subscription Status

Yes 27%

No 73%

Revenue by Category

Sales by Category

Revenue by Age Group

Sales by Age Group

Donut chart showing subscription status: Yes (27%) and No (73%)

Bar chart showing revenue by category: Clothing (90K), Accessories (60K), Footwear (30K), Outerwear (15K)

Bar chart showing sales by category: Clothing (1800), Accessories (1200), Footwear (600), Outerwear (300)

Horizontal bar chart showing revenue by age group: Young Adult (55K), Middle-aged (50K), Adult (45K), Senior (40K)

Horizontal bar chart showing sales by age group: Young Adult (1000), Middle-aged (900), Senior (800), Adult (700)

6. Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers.

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control.

Product Positioning

Highlight top-rated and best-selling products in campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.